TP3 Report: Generative Models (VAE and GAN)

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1 Part 1: Variational Autoencoder

Our VAE (latent dimension 20, 652,824 parameters) trained without any problem. The ELBO loss per sample went from 260.70 at epoch 1 to 240.94 at epoch 15 and decreased at every epoch.

Question 1.1. *Compare the reconstructions to the random prior samples. Which are sharper, and why?*

The reconstructions are clearly sharper than the random samples. They keep the right class and the global shape of each item (the ankle boot, the trousers, the pullover), but small details are lost: the "Lee" logo and the checked pattern of the shirt disappear. The prior samples are even more blurry, and some of them look like a mix of two items, for example a top merged with trousers.

This makes sense. For a reconstruction, the encoder has seen the real image, so μ already contains information about that particular item. For a prior sample, z is drawn from $\mathcal{N}(0, I)$ with no image behind it, so the decoder has to guess a plausible item and ends up producing an average of several possible items.

Question 1.2. *Is the interpolation smooth or abrupt? What does it say about the effect of the KL term?*

The interpolation is smooth. Going from the ankle boot to the pullover, the boot fades out while the sleeves and the body of the pullover appear. There is no sudden jump and no image that turns into pure noise. The images in the middle are simply a blurry mix of the two shapes.

This shows that the latent space is continuous. The KL term pushes every $q(z|x)$ toward the same distribution $\mathcal{N}(0, I)$, so all the codes are grouped in one region around the origin without empty gaps between them. Because of this, every point on the line between two codes still decodes to something meaningful.

Question 1.3. *What happens to reconstruction sharpness with $\beta > 1$?*

The reconstructions become more blurry. The loss has two terms that work against each other: the reconstruction term wants z to keep as many details of x as possible, while the KL term wants $q(z|x)$ to stay close to $\mathcal{N}(0, I)$, which limits how much information z can carry. With $\beta > 1$, the KL term has more weight, so z keeps less information about each image. In exchange, the latent space becomes more regular.

2 Part 2: Minimax and non-saturating generator losses

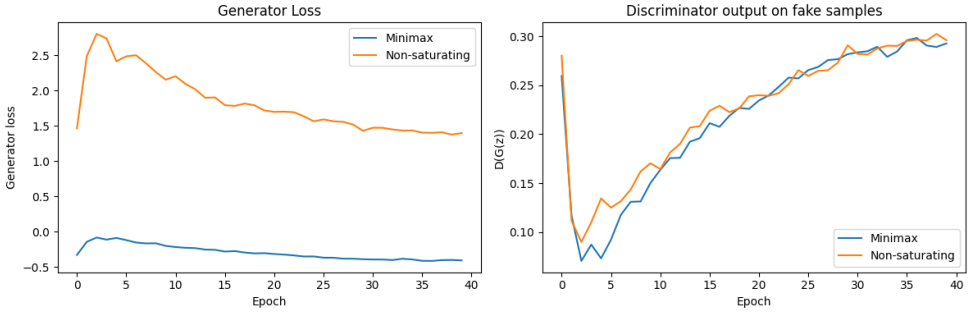


Figure 1: Generator loss (left) and $D(G(z))$ (right) for both runs, with the same seed and hyperparameters.

Question 2.1. *Describe the generator loss and $D(G(z))$ curves and explain their behaviour.*

The two generator loss curves cannot be compared directly, because the two formulas do not have the same scale. The minimax loss $\log(1 - D(G(z)))$ is always negative and almost flat (from -0.33 to -0.41). The non-saturating loss $-\log D(G(z))$ goes up to 2.80 at epoch 3 and then slowly decreases to 1.40.

The $D(G(z))$ curve is more useful, since it means the same thing in both runs. In both cases it first drops quickly (to 0.071 for minimax and 0.090 for non-saturating at epoch 3), because the discriminator learns faster than the generator at the beginning. After that, it rises steadily to about 0.29 at epoch 40 (0.293 and 0.296). The discriminator loss also goes from about 0.5 to 1.0, getting closer to the equilibrium value $2 \log 2 \approx 1.39$. So the generator is catching up, but training is not finished after 40 epochs.

The difference between the two losses is in the gradient. If f is the logit of D, so that $D = \sigma(f)$, the derivative of the generator loss with respect to f is $-D(G(z))$ for minimax and $-(1 - D(G(z)))$ for non-saturating. When $D(G(z))$ is small, the minimax gradient becomes very small. At epoch 3 it is about 0.07, against 0.93 for the non-saturating loss, so around 13 times weaker. This is the vanishing gradient problem seen in CM5.

In our run, however, this did not block the minimax generator, and the two $D(G(z))$ curves are almost on top of

each other. We see two reasons for this. First, $D(G(z))$ never reached 0 (its minimum was 0.07), so the gradient was small but not zero. Second, Adam divides each update by an estimate of the gradient size, so a gradient that is smaller everywhere still gives updates of similar size. With a stronger or faster discriminator, $D(G(z))$ would get closer to 0 and the minimax version would probably get stuck.

Question 2.2. *Does the run with the flatter loss curve also give worse or less diverse samples?*

No. The two sample grids have similar quality. In both, we can recognize sneakers, ankle boots, shirts, pullovers and dresses, with sharp edges but a lot of noise inside the shapes. Neither run shows mode collapse, since several different classes appear in each grid. So the minimax run, which has the flatter loss curve, did not produce worse samples here. This shows that the value of the generator loss is not a good indicator of sample quality.

Question 2.3. *Why is the non-saturating loss used in practice if the minimax optimum is the true data distribution?*

Goodfellow et al. prove that the optimum of the minimax game is the true data distribution, but this says nothing about how easy it is to reach it with gradient descent. At the start of training, the discriminator easily rejects the fake images, so $D(G(z)) \approx 0$ and the minimax gradient almost disappears. The non-saturating loss has the same optimum but gives a strong gradient exactly when the generator is bad, which is why it is preferred in practice.

3 Part 3: Comparison of VAE and GAN

Question 3.1. *Comparison table.*

Dimension	**VAE**	**GAN**
Training stability	Very stable: the loss decreases at every epoch, from 260.70 to 240.94.	Less stable: $D(G(z))$ falls to about 0.08 at epoch 3 before recovering, and the non-saturating loss first jumps from 1.47 to 2.80.
Sample sharpness	Blurry: details like the "Lee" logo are lost, and prior samples are smooth and sometimes mix two items.	Sharper edges and more contrast, but a lot of noise inside the items in both runs.
Latent space structure	Continuous and usable: the interpolation from boot to pullover is smooth, and real images can be encoded.	There is no encoder and nothing in the loss organizes z, so we cannot encode an image or control what is generated.
Does the loss track quality?	Yes: a lower ELBO loss goes with better reconstructions.	No: the minimax loss is almost flat (-0.33 to -0.41), yet its samples are as good as the non-saturating ones.

Question 3.2. *Which model would you choose if the loss had to be used as a stopping criterion?*

We would choose the VAE. Its loss is a single objective minimized by gradient descent, and in our run it decreased smoothly together with the improvement of the reconstructions, so we can stop training when it stops decreasing. The GAN loss depends on the current discriminator: it can change because D changed and not because the images improved, so we still need to look at the samples.

Question 3.3. *Which failure mode is more acceptable in an application of your choice?*

Suppose we want to generate extra images to train a clothing classifier. In this case, blur is more acceptable than mode collapse. A blurry boot still has the right shape and class, and the VAE covers all the classes of the dataset. Mode collapse would remove some classes completely, and the classifier would never learn them.

Question 3.4. *Why does the VAE give blur and the GAN give noisy images with similar MLPs?*

The difference comes from the training signal. The VAE decoder is trained with a pixel by pixel BCE against one target image. When several details are possible for the same z, the best choice for this loss is their average, which gives grey and blurry pixels. The GAN generator never sees a target image and only tries to fool the discriminator. Since real items have sharp edges, blurry images are easy to detect, so the generator produces strong contrasts. But nothing forces each pixel to match a real image, so the result contains noisy textures.